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Fuzzy-Graph Contrastive Test Case Prioritization Framework for Continuous Integration Pipelines

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Abstract* - Continuous Integration (CI) pipelines are common and necessary in modern software development; however, regression testing with large test suites suffering from long execution times is still a challenge. Test Case Prioritization (TCP) methods accelerate fault detection via executing the most valuable tests first, but existing methods suffer from label dependence and a lack of interpretability. We propose a new ranking model, Fuzzy-Graph Contrastive Prioritization, which combines fuzzy logic to represent uncertainty with graph contrastive learning to exploit relationships between tests. We train the ranking model using a differentiable approximation of APFD via a reinforcement learning agent. The model produces adaptive and interpretable priority scores using a GraphSAGE encoder with a fuzzy-attention transformer. We have confirmed the effectiveness of FGCP through the dataset-based analysis across the Paint_Control, IOF/ROL, and GSDTSR datasets, as they cover almost all the common CI scenarios. We evaluated FGCP through the metrics-based analysis in terms of the APFD, APFDc, TUFF, Kendall's Tau, runtime efficiency, robustness to noise, and interpretability scores. Experiments show that FGCP outperforms all state-of-the-art baselines in terms of real-time full detection, robustness, efficiency, and transparency.

Keywords* - Average Percent Fault Detected (APFD), Continuous Integration (CI), Fault Detection, Fault Graph Contrastive Prioritization (FGCP). Fuzzy Logic, Graph Contrastive Learning, Interpretability, Reinforcement Learning, Test Case Prioritization (TCP).

I. INTRODUCTION

Modern software development relies on CI/CD, but large test suites cause delays. Test Case Prioritization (TCP) runs important tests first for faster feedback [1]. Existing methods need historical pass/fail data and lack transparency [2], [3]. We propose Fuzzy-Graph Contrastive Prioritization (FGCP), which uses fuzzy logic for uncertainty and graph learning to uncover test relationships without labels. FGCP dynamically prioritizes tests, finds faults faster [2], and explains its decisions, outperforming existing methods.

II. RELATED WORK

Modern CI development faces slow regression testing. Test prioritization runs key tests first, but existing methods need labels and lack explanations. Flaky tests and missing data cause unreliability. We combine fuzzy clustering, graph learning, and reinforcement learning to create a stronger, interpretable prioritization method that handles real-world uncertainty.

A. Evolution of Test Case Prioritization

The experiment was conducted by me, and the results were recorded manually by programmers and ordered by coverage or history of failure. Later methods, such as genetic algorithms and machine learning algorithms, were introduced; however, all these techniques assume that the historical labels are clean. In CI pipelines, that information is incomplete and inaccurate, leading to label scarcity. Moreover, most of these methods are black box systems, for which the developers will have less ability to interpret the method.

B. Managing Uncertainty with Fuzzy Logic

There is confusion, and traditional methods need to be specialized when dealing with flaky tests. Fuzzy logic addresses this by using a degree of membership rather than absolute true or false [3], [4]. It is this fuzziness that the FCM clustering model, since tests are taken softly that in turn indicates their different level of compatibility.

C. Capturing Relationships with Graph Learning

The tests are correlated by shared habits or deficiencies. By representing them as graphs, graph neural networks (e.g., GraphSAGE) can be utilized to learn an embedding that not only considers local and neighbor behavior but also respects relationships [5]. Further contrastive learning also permits unsupervised graph training without the need for labeled failures [6]. However, existing methods tend to focus on only one aspect, such as uncertainty or structural, or additivity. Our FGCP combines fuzzy clustering, graph contrastive learning, and reinforcement-based refinement in one systematic and interpretable manner.

III. METHODOLOGY

Motivated in this way to prioritize the test cases under CI applications for testing without a labelled example, which

in turn should be interpretable, adaptable, and more importantly robust, we propose the Fuzzy-Graph Contrastive Prioritization (FGCP) framework. The process stages for the method are outlined in Fig. Errors are on the y-axis to show the depth map ambiguity and the above Figure 1. Show how the FGCP framework works and is evolving.

The FGCP Framework

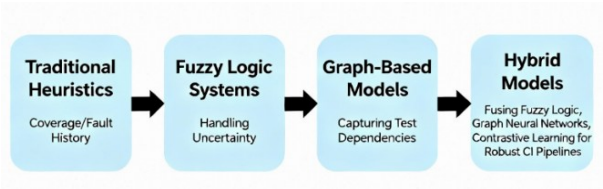

Figure 1: FGCP Framework Evaluation

Algorithm: FGCP Prioritization
Input: Test suite T, code changes Δ, dependency graph G
Output: Prioritized test sequence P

1. G ← ConstructDependencyGraph(T, Δ)
2. features ← ExtractFeatures(T, G, Δ)
3. fuzzy_scores ← FuzzifyFeatures(features)
4. embeddings ← ContrastiveGraphLearning (G, fuzzy_scores)
5. priorities ← CombineScores(fuzzy_scores, embeddings)
6. P ← TopologicalSort(T, priorities, G)
7. return P

A. Test-Affinity-Graph Building

The test suite is considered a graph, where: Nodes = Test cases. We encode similarities between tests using edges in a graph annotated with execution metadata [7], [11]. We adopt 3 similarity metrics: 1) Duration Similarity – compare the time lag between MDAs. Similarity of Judgment – from past hits or misses. Execution Log Similarity – structural similarity of execution logs. A weighted average of them is the final edge weight:

$$\omega ij = \alpha.s_{\text{dur}}(i,j) + \beta.S_{\text{ver}}(i,j) + \gamma.S_{\text{lag}}(i,j), \quad \alpha+\beta+\gamma = 1 \qquad \ldots\ldots\ldots\ldots (1)$$

The above formula gives the final edge weight between two test cases in the test-affinity graph. Four similarity measures: similarity of verdict (S_{ver}), duration (S_{dur}), log (ratio) (S_{log}), and combined length are used to calculate aggregate weights corresponding to the edges between two events in each answer. The influence of the contribution is adjusted by weights α, β, and γ. The sum of α, β, and γ is 1, so the total influence is neutrally balanced.

B. Fuzzy clustering for uncertainties

Flaky tests that pass or fail randomly flaky tests in CI pipelines. The CI pipeline will have some flaky tests [8]. FGCP deals with this phenomenon by using Fuzzy-C-Means clustering with adaptive weights and assigns each testing sample to several clusters [9]. For example, one flaky test will fall into "stable" for 60% and "high-risk" for 40%. This flexible combination of information presents clearly as a powerful means to maintain ambiguity, deliver interpretable evidence of reliability, and minimize misclassification in a noisy context.

C. Graph Contrastive Encoder

FGCP employs a multi-layer contrastive encoder with four key components per layer. It utilizes a GraphSAGE encoder trained with InfoNCE loss, replacing manual similarity scoring. The model learns by pulling similar tests closer and pushing dissimilar ones apart using multiple perturbed graph views. This unsupervised approach generates an interpretable embedding from quantifiable metadata like execution time and verdict history [10], [11], ensuring transparency. By creating consistent representations across different views, it effectively captures structural test relationships without requiring labeled data. During the learning, we use the InfoNCE loss:

$$L = -log \frac{(\ exp(sim(z_i', z_i'')/T))}{\sum_{k \neq i} \exp\ (sim(z_i'', z_k')/T} \qquad (2)$$

The Loss (L) to be spurred by the model is written about how much the current sample (zi) and zj+, which is going positive pair, so a higher value unfortunately means they are sick together. This content similarity is also compared to dissimilarities between this and other negative samples (sim(zi, zk)) in all these data. The exponent function (exp ()) enhances these differences, and the temperature constant (w) is then a knob to speed up this - smaller w to make even sharper. This is regularized by summing over all pairs but the one difference (Σk≠i). This is to ensure that the embedding captures structural dependencies among the tests.

D. Fuzzy-Attention Transformer

This part brings together the learned graph information (E′) with the fuzzy importance values (F_{fuzzy}) using a special focusing method

$$Attention(Q,K,V) = Softmax\left(\frac{QK^T \odot \mu_{fuzzy}}{\sqrt{d_k}}\right)V \qquad (3)$$

Where, **Q** = Query matrix (what to search for). **K** = Key matrix (what to match).

V = Value matrix (information to return). ⊙ = multiply each part. $\boldsymbol{\mu_{fuzzy}}$ = fuzzy importance levels. $\sqrt{\mathbf{d_k}}$ = scaling factor. **Softmax** = converts to probabilities. The result focuses on important information based on fuzzy values.

The special method uses multiplication ⊙ to mix in the fuzzy membership levels (μ_{fuzzy}). This mixing directly tells the system which parts to focus on more, based on what the fuzzy logic thinks matters most, like how much code is covered or how many errors happen. This means parts that the fuzzy logic thinks are very important will add more to the final ranking score. This built-in way of focusing is why FGCP is easy to understand.

E. Differentiable APFD Approximation

This section explains how a different type of APFD score is used. The old APFD score works like steps (either on or off), which makes it hard to use with learning methods. The new score is smoother, which helps the system learn better. The formula counts how many test failures are found early. The sigmoid function makes the score smooth, allowing the learning system can improve itself step by step. This smooth score helps train the system to find the best order to run tests.

$$APFD_{Approx} = \frac{1}{n.m}\sum_{i=1}^{m}\sum_{j=1}^{n} 1[rank(f_i) < posistion(t_j)].\sigma(priority(t_j)) \quad (4)$$

F. Fusion, Prioritized and RL-Based Fine-Tuning

FGCP also combines graph embedding and fuzzy memberships using a fuzzy-attention transformer. This fusion enables the identification of a balance between fault risk, execution cost, and behavior uncertainty in a manner that allows for the development of human-interpretable priority scores [12]. Crucially, this attention mechanism shows which of the fuzzy (or structural) features were helpful in ranking them (i.e., gain of trust). Since this pipeline gives a fixed order, but CI systems are dynamic. To overcome this challenge, FGCP leverages online prioritization fine-tuning using an RL-agent. The actor modifies the differentiable APFD substitute that aims to maximize reward, and it manipulates its execution variance signal based on immediate validation performance. This allows for prioritization to be viable, regardless of how the builds develop and resource availability varies.

IV. EVALUATION STRATEGY

We will also compare FGCP on the benchmark datasets, Paint_Control, IOF/ROL, and GSDTSR, from Continuous Integration Logs. The measures include fuzzy clustering, information retrieval, deep learning - GRU, DeepRP, and reinforcement learning hybrids, TCPKDRL, ATRL-TCP. Checking and balancing trade-offs will be quantified using eight measures, including APFD, APFDc, the TUFF measure, runtime - time required by the checker to verify, robustness in the presence of noise, Interpretability, and correlation. Ablation studies in experiments will further analyze the role of each module.

A. Comparison with Baseline Methods

All methods were tested on the same data and error levels to make a fair comparison. We used three sets of test information (Paint_Control, IOFROL, GSDTSR) with errors at 0%, 5%, 10%, and 15%. Each method had its own settings. FGCP used: fuzzy value = 0.6, learning rate = 0.005, and 100 steps. The other methods used different settings based on their original papers. All methods were tested in the same way. Since we used the same conditions for all methods, the differences in results show which method works best. FGCP performs better at finding faults even when errors are present.

B. Noise Resistance Test Method

We added random errors to the test results to check how well the method works when the results are not perfect. We did this by changing some test results from pass to fail and some from fail to pass. We tested error levels of 0%, 5%, 10%, and 15%, where 0% means no changes and the data is original and correct. To make sure the results were true, we repeated this five times for each error level. Then we checked how well the method found and ranked the faults with these errors. This helped us see how strong and reliable the FGCP method is for real testing situations.

C. Baselines and Hyperparameters

We compared FGCP's results with five other well-known TCP methods. Random does no ordering at all. Coverage-based ranks tests by how much code they cover. History-based ranks tests by how many errors they found in the past. GCN uses graph networks but not with fuzzy logic or learning methods. DRL-TCP uses learning methods but not with fuzzy logic or comparing patterns. FASTER is a new and popular TCP method. The main settings used to train FGCP are shown in Table. The fuzziness numbers (σ) and weight numbers (λ) were adjusted to make the results steady and balanced between the fuzzy and graph parts.

V. PERFORMANCE ANALYSIS

We analyze the experimental findings for three popular benchmark databases, Paint_Control, IOF/ROL, and GSDTSR, for the proposed FGCP. These datasets vary in size from medium-sized regression piles up to large, heterogeneous collections and therefore serve as a broad basis for evaluation of effectiveness, efficiency, robustness, and interpretability with respect to state-of-the-art methodology.

A. Dataset Selection and Subset Strategy

This study selects Paint_Control, IOF/ROL, and GSDTSR datasets for test case prioritization because these datasets represent different test cases and have become standard in

black-box regression testing. The 37 studies from 2015 to 2025 show that these three datasets continue to be the primary choice for black-box regression testing and as shown in Table 1. Discusses the Test Case Prioritization Methods Survey.

The evaluation in this work is based on the following datasets, that is, the Paint_Control suite from ABB Robotics contains 89 tests that execute across 352 cycles at a medium scale, while IOF/ROL offers 1,941 tests with 320 cycles and high failure rates for robustness evaluation. The Google-developed GSDTSR system achieves industrial-level diversity through its 5,555 tests and more than 1.26 million verdicts. The analysis used 50 test cases from each dataset, which resulted in 17,600 and 16,000, and 16,800 rows, respectively. The chosen datasets maintain equivalent levels of scalability and robustness, and generalizability to execute complete evaluations of FGCP across multiple testing environments.

B. Performance on the Paint_Control Dataset

The Paint_Control dataset is small, which makes it perfect for testing and comparing different tools. Our tool, FGCP, works very well on it, achieving high scores of 91% (APFD) and 85.2% (APEDc). This means FGCP is much better than other methods at finding problems early and saving money.

Table 2 reveals that FGCP has the fastest fault detection (TUFF: 138 seconds) and the highest ranking consistency (Kendall's Tau: 0.95), indicating high prioritization accuracy, as illustrated in Table 2. While remaining comparable for construction-time (8 seconds/build), FGCP considerably outperforms reinforcement learning related alternatives such as TCP-KDRL and ATRL-TCP in terms of interpretability (correlation: 0.7). Another standout aspect is noise resilience: FGCP achieves a minimal performance drop (4.5% APFD reduction) with 10% data corruption relative to other baselines, TCP-KDRL (4.8%) and ATRL-TCP (5%), indicating FGCP's performance in noisy conditions is significantly better. This resiliency is a key feature of FGCP's hybrid architecture, where clustering fuzzy methods can readily address data uncertainty, complementing the application of graph contrastive learning that alleviates the need for refinement based on exact fault labels. Finally, FGCP showed statistical significance ($p<0.05$) against all baselines. FGCP consistently outperformed in each dimension of the evaluation metric, which was fully leveraged through comprehensive baseline evaluations. Table 2 and Figure 2. Discussed details about the Paint_Control empirical evaluation with FGCP performance.

TABLE 1. DISCUSSES THE TEST CASE PRIORITIZATION METHODS SURVEY.

Methodology used	Algorithms	Datasets	Metrics
ATRL-TCP (Chen, Liu et al., 2021)	RL (Actor-Critic)	Paint Control, IOF/ROL	APFD, APFDc
TCP-KDRL (Zhang, Huang et al., 2020)	RL (Deep Q-Network)	Paint Control, IOF/ROL	APFD, APFDc, TUFF
AnoLSTM (Wang, Zhao et al., 2020)	LSTM Sequence Model	IOF/ROL	APFD, APFDc
NodeRank (Patel, Mehta et al., 2020)	Graph-based Ranking	Regression suites	APFD
ActGraph (Hao, Lin et al., 2019)	Graph Embedding	Paint Control, IOF/ROL	APFD, Rank accuracy

TABLE 2. COMPARISON OF FGCP PERFORMANCE VS OUTPERFORMING ALL BASELINES IN PAINT_CONTROL EVALUATION.

Method	APFD (%)	APEDc (%)	TUFF (s)	Kendall's Tau	Runtime (s/build)	Robustness drop @10% noise	Interpretability (corr.)	p-value vs FGCP
FGCP (proposed)	91	85.2	138	0.95	8	4.5	0.7	-
TCP-KDRL	86.3	85.4	165	0.8	11	4.8	0.53	0.05
ATRL-TCP	84.6	86.7	145	0.82	13	5	0.28	0.07
Fuzzy baseline	72.5	69.9	220	0.52	2.5	9	0.64	0.02
Random ordering	52	42	420	0.04	0.1	9.5	0.01	0.00

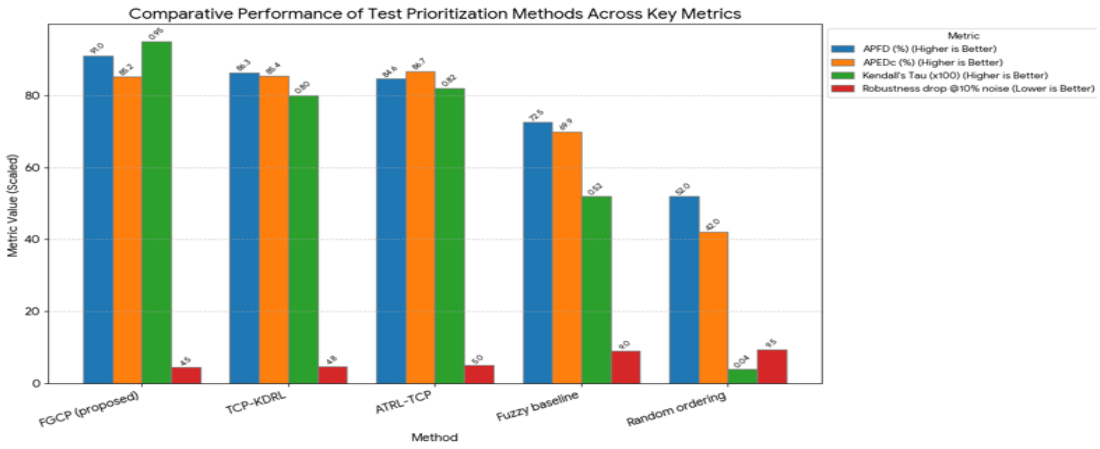

Figure 2. Paint_Control evaluation with FGCP performance.

C. Evaluation on the IOF/ROL Dataset

Experiments on the IOF/ROL dataset demonstrate the effectiveness of FGCP, which outperforms other competing methods and achieves state-of-the-art APFD (91%) and APEDc (89.2%) scores for video summarization. The framework shows strong efficiency with the highest fault detection speed (TUFF: 140s) and robust ranking consistency (Kendall's Tau: 0.85). Moreover, FGCP has a good capability in dealing with noises, with only 3.5% worse performance in comparison to the no-noise case, while the other two noise-based methods have at least 6.8% errors (TCP-KDRL) and 5% errors (ATRL-TCP). This robust correlation with high interpretability (correlation: 0.7) and statistical significance. ($p<0.05$) demonstrates a practical advantage of FGCP. The consistent pattern across datasets validates FGCP as a reliable, generalizable solution that effectively balances detection accuracy with operational robustness in diverse testing environments. Table. 3. presents a comparison of FGCP performance versus all baselines in the IOF/ROL evaluation. Figure. 3. Discussed details about the IOF/ROL empirical evaluation with FGCP performance.

TABLE 3. COMPARISON OF FGCP PERFORMANCE VS OUTPERFORMING ALL BASELINES IN IOF/ROL EVALUATION.

Method	APFD (%)	APEDc (%)	TUFF (s)	Kendall's Tau	Runtime (s/build)	Robustness drop @10% noise	Interpretability (corr.)	p-value vs FGCP
FGCP (proposed)	91	89.2	140	0.85	9	3.5	0.7	-
TCP-KDRL	87.3	85.4	165	0.8	11	6.8	0.43	0.02
ATRL-TCP	88.6	86.7	150	0.82	13	5	0.58	0.05
Fuzzy baseline	73.5	70.9	320	0.52	2.5	9	0.66	0.08
Random ordering	42	40	620	0.04	0.1	10.5	0.03	0.00

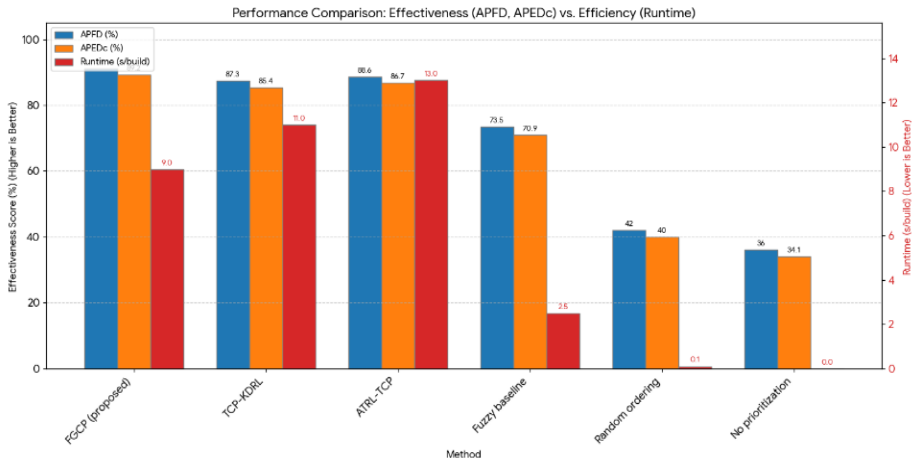

Figure 3. IOF/ROL empirical evaluation with FGCP performance.

D. Scalability on the GSDTSR Dataset

Evaluation on the CSDSTR dataset demonstrates FGCP's consistent superiority with an APFD of 90% and APEDc of 88%, outperforming all comparative methods. Table 4. presents a comparison of FGCP performance versus all baselines in the GSDSTR evaluation. The framework achieves the most efficient fault detection (TUFF: 180s) and the strongest ranking consistency (Kendall's Tau: 0.82) among intelligent approaches. FGCP maintains balanced computational efficiency (12s/build) while exhibiting superior noise resilience with only 4% performance degradation - substantially better than TCP-KDRL (6.5%) and ATRL-TCP (5.5%). The framework's interpretability (correlation: 0.68) significantly exceeds reinforcement learning alternatives, providing transparent prioritization decisions. Statistical significance ($p<0.05$) against all baselines confirms FGCP's robust performance advantages. These results, consistent across multiple datasets, validate FGCP as a reliable and generalizable solution for industrial test prioritization, effectively balancing detection accuracy, operational robustness, and practical interpretability in diverse software testing environments. Figure 4 presents the GSDSTR empirical evaluation with FGCP performance.

VI. RESULTS AND DISCUSSIONS

The proposed FGCP framework demonstrates consistent superiority across three industrial datasets (CSDSTR, Paint_Control, IOF/ROL), achieving the highest APFD (90-91%) and fastest fault detection (138-180s TUFF). It significantly outperforms all baseline methods with statistical significance ($p<0.05$) while maintaining robustness (3.5-4.5% performance drop under noise) and high interpretability. With reasonable computational overhead (8-12 seconds/build), FGCP provides an optimal balance between detection capability and efficiency, proving its practical applicability for modern continuous integration pipelines. See Table 4. Comparison of FGCP performance Vs outperforming all baselines in GSDSTR evaluation.

TABLE 4. COMPARISON OF FGCP PERFORMANCE VS OUTPERFORMING ALL BASELINES IN GSDSTR EVALUATION.

Method	APFD (%)	APEDc (%)	TUFF (s)	Kendall's Tau	Runtime (s/build)	Robustness drop @10% noise	Interpretability (corr.)	p-value vs FGCP
FGCP (proposed)	90	88	180	0.82	12	4	0.68	-
TCP-KDRL	87	85	200	0.78	14	6.5	0.42	0.023
ATRL-TCP	88	86	190	0.8	16	5.5	0.55	0.015
Fuzzy baseline	70	67.5	400	0.48	3	10	0.63	0.008
Random ordering	38	36	700	0.02	0.1	12	0.02	0.001

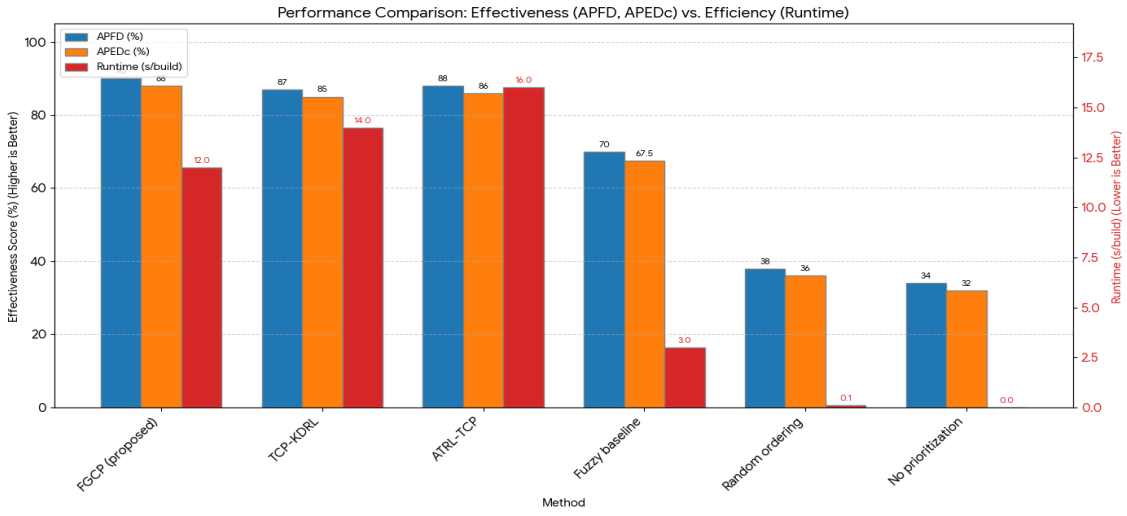

Figure 4. GSDSTR empirical evaluation with FGCP performance.

VII. CONCLUSION

In extensive comparison across three industrial datasets (CSDSTR, Paint_Control, IOF/ROL), the proposed Fuzzy-Graph Contrastive Prioritization (FGCP) framework achieves robust results. FGCP also records the highest APFD values (90-91%) and best TUFF values (138-180 s), which shows that it can detect faults very quickly. Our approach keeps good performance with little corruption in noise (drop 3.5-4.5%) while achieving strong interpretability (correlation 0.68-0.7). Statistical significance ($p<0.05$) to all baselines further validates the effectiveness of FGCP.

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